Topic 2.5 - Exponential Function Context and Data Modeling

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: ____________ Class: ____________ Date: ____________

Why this topic matters

Exponential modeling begins with a multiplicative mechanism, not merely a curved graph. A strong model identifies the reference value, the factor and its interval, the input domain, and the assumptions that make repeated proportional change reasonable.

Learning targets

- Construct exponential models from percent rates, two points, transformations, or regression output.
- Use exponential models to predict outputs and solve for inputs.
- Interpret parameters, equivalent time scales, domains, units, and model limitations.

Language and structure

Term	Meaning in this topic
growth factor	The multiplier greater than 1 applied over a specified interval.
decay factor	The multiplier between 0 and 1 applied over a specified interval.
doubling time	The input increase required for a model output to multiply by 2.
half-life	The input increase required for a model output to multiply by $1/2$.
extrapolation	Using a model beyond the observed or intended input interval.

Launch: make a claim before calculating

Launch investigation

Reasoning first

A quantity is measured as 50, 61, 74.4, and 90.8 at equally spaced times. Decide whether an exponential model is plausible, estimate the factor, and name one additional check needed before using the model for prediction.

Concept 1: Build from an initial value and rate

Core idea

A percent increase of r gives factor $1 + r$; a percent decrease of r gives factor $1 - r$.

- Use $P(t) = P_0 b^t$ when P_0 is the value at $t = 0$ and b is the factor per input unit.
- If the stated factor applies every k units, use $P(t) = P_0 R^{t/k}$.
- Units should appear in the interpretation even though the factor itself is dimensionless.

Worked example - annotate the reasoning

Problem. A culture starts with 1200 cells and grows by 18% each week. Write a model, predict the population after 7 weeks, and find when it first exceeds 5000 cells.
Model reasoning. $P(t) = 1200(1.18)^t$. $P(7) \approx 3822.57$. Solving $1200(1.18)^t > 5000$ gives $t > 8.62$, so the first whole week is 9.

Check your understanding

Independent

A device worth 4800 QAR loses 14% of its value each year. Write the model and find its value after 5 years.

Concept 2: Build from two points or a characteristic time

Core idea

For $P(t) = ab^t$, the ratio of two outputs determines b after accounting for the input gap.

- $b = \left(\dfrac{P(t_2)}{P(t_1)}\right)^{1/(t_2 - t_1)}$.
- Doubling time d gives $P(t) = P_0 2^{t/d}$.
- Half-life h gives $P(t) = P_0 (1/2)^{t/h}$.

Worked example - annotate the reasoning

Problem. An exponential model satisfies $P(2) = 360$ and $P(6) = 5760$. Determine the model $P(t) = ab^t$.
Model reasoning. $5760/360 = 16$, so $b^4 = 16$ and $b = 2$. Then $360 = a(2)^2$, giving $a = 90$. Therefore $P(t) = 90(2)^t$.

Check your understanding

Independent

A substance begins at 640 mg and has half-life 9 hours. Find the amount after 24 hours.

Concept 3: Shifted exponential models and limitations

Core idea

A model $P(t) = L + Ab^t$ approaches the limiting value L rather than zero.

- Subtract the limiting value before finding ratios from data.
- Interpret $A = P(0) - L$ as the initial difference from the limit.
- Model fit does not establish causation, and a good short-term fit may fail under long-term extrapolation.

Worked example - annotate the reasoning

Problem. A temperature model approaches $20°$C and satisfies $T(0) = 95$ and $T(3) = 58.4$. Find $T(t)$.
Model reasoning. Let $T(t) = 20 + Ab^t$. From $T(0) = 95$, $A = 75$. Then $58.4 - 20 = 38.4 = 75b^3$, so $b^3 = 0.512$ and $b = 0.8$. Thus $T(t) = 20 + 75(0.8)^t$.

Check your understanding

Independent

Why is it unreasonable to extend a bacterial growth model indefinitely in a closed container?

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

Write a model for an initial amount of 300 that decreases by 9% per day.

Exit ticket 2

No calculator

A model doubles every 5 hours. By what factor does it change in 15 hours?

Exit ticket 3

No calculator

Interpret the parameter L in $P(t) = L + A(0.7)^t$.